BLACK COLLATERAL

AN INTELLIGENT STORE OF VALUE TO BET AGAINST BITCOIN

Dec 2020

V2.2

ABSTRACT

This Whitepaper introduces a new innovative high store of value cryptocurrency called BLACK COLLATERAL (BLACK), where users are restricted from selling the tokens in large quantities using a novel time lock mechanism. As a result, the investors are forced to hold the currencies for an extended period that fundamentally makes the economic model different from other cryptocurrencies. Additionally, Black is backed by a pool of crypto assets locked in a crypto reserve called Burn Vault to provide minimum guarantee price support, where the user can cash out without impacting the market price.

Disclaimer: Before the complete implementation of the project, the logic may change without notice. This white paper documents the dynamics while seeking to provide details on key areas of importance.

1. INTRODUCTION

1.1 THE MAJOR CRYPTO MARKET PROBLEM

Synthetic commodity money is a new classification for scarce assets (like GOLD, SILVER, AND DIAMOND). Bitcoin is an example of Synthetic commodity money that could reform money and be a better alternative to traditional assets. Gold is one of the best store of value for its unique properties, interwoven into cultures for thousands of years. People continued to hold gold for various reasons throughout the centuries. Modern economies have placed value on gold, thus perpetuating its worth. The concept of money is changing drastically, with cryptocurrency revolutionizing the world as a vital forefront runner.

The birth of bitcoin solves many financial trust issues and, at the same time, introduces more problems. One of the critical crypto market investment risks we are facing nowadays is that the value of almost all crypto projects is pegged to bitcoin. As a result, the entire market comes down when bitcoin goes down. High correlations in the decentralized market prohibit diversification due to several price fluctuations that affect all other cryptocurrencies in general.

The main factor for such price fluctuations is the lack of adequate control when big investors decide to cash out in bulk, which completely crashes the token price. Innovative experiments to create other forms of money to bet against bitcoin have been going for so long that computer scientists introduced several protocols to minimize price fluctuations and de-correlate market dependencies.

In theory, to design such price resistant crypto-asset models,

1) There must exist a mechanism where significant large investors' cashouts are controlled to preserve small investors.
2) There also must exist a mechanism where investors can exchange crypto assets for FIAT or other stable assets without impacting the market price.

1.2 TARGET AND SOLUTIONS

Since 2018, Project BLACK designed several models to define an innovative risk-free, price-resistant investment class for the DeFi economy that has the following two fundamental characteristics

A) Minimal price volatility footprint resisting bitcoin corrections

B) Growing Purchase Power

For example, suppose an investor buys $1000 worth of an asset class as a long-term investment, his/her purchase power gradually grows like a plant always, irrespective of the crypto market position, bitcoin price, or any other economic shocks.

After years of research, we are proud to introduce a new digital asset class called BLACK COLLATERAL (BLACK), an intelligent store of value especially designed to bet against bitcoin by solving several cryptocurrency market corrections altogether.

2. BLACK COLLATERAL – A BETTER ALTERNATIVE TO GOLD

BLACK aims to be the world's first NON-DEPRECIATING Smart Money with all the characteristics of GOLD by introducing three innovative protocol operations that help increase the purchasing power over time significantly

A. **TIME-LOCK MECHANISM** is a novel design that prevents investors from selling their assets in large quantities. This mechanism regulates purchase power by enforcing a time-lock on the holder's wallet, due to which one can't sell or transfer more than 1% of the asset in 24 hours. By enabling the time-lock, asset holders, especially whales, can't manipulate the price significantly, resulting in a long-regulated purchase power increase.

B. **BURN VAULT MECHANISM** is a new decentralized reserve pool similar to the Bretton Woods FIAT-Gold Reserve system that provides a minimum collateral price guarantee to the investors irrespective of the market condition during the panic period.

C. **AUTOMATED LIQUIDITY POOL (ALP) MECHANISM** prevents exchange liquidity fluctuation to a greater extent by locking a portion of every transaction back in the liquidity pool, enhancing the overall market price of the crypto asset.

In summary, BLACK is designed to be precious in every way because it is scarce, non-dilutive, free from market distortions, resistant to economic shocks, and most importantly, backed by another crypto asset, increasing integrity and trust.

3. WHAT PROBLEMS DOES BLACK SOLVES?

A. DIGITAL ASSET INTRA-CORRELATION

Since the inception of Bitcoin in 2009, dozens of other cryptocurrencies (and synthetic commodities) have spun off, representing a financial value on an electronic, data-driven ledger of some sort. Most synthetic commodities' price and weight depend highly on the price of Bitcoin, leading to somewhat of a homogeneous system of synthetic commodities. Ideally, a digital asset would exist with almost no correlation to traditional asset groups and existing digital assets. That's where Black Collateral comes in. The entire ethos of this project is a paradigm shift from price-based trading strategies to time-based trading strategies.

B. TIME-BOUND BUY-SELL RATIO

BLACK controls the time-demand equilibrium every 24 hours using the Time-Lock mechanism tied to the number of tokens each investor holds in their wallets. Time-Lock ensures that the market cap of the collateral persists and the growth is consistent by adjusting the selling threshold dynamically.

C. DIVERSIFICATION OF CRYPTO PORTFOLIOS

Since BLACK is resistant to crypto corrections, it offers an entirely different model to minimize portfolio classification risks, making it one of the best low-risk strategies for long-term asset holders who want to bet against bitcoin.

D. INFLATION HEDGE

BLACK will be an excellent hedge against inflation because its price will rise over time as the supply decreases. When other crypto assets lose their purchasing power to inflation, Black tends to be priced in those currency units and tends to rise along with everything else. Moreover, Black will be a good store of value, so people may be encouraged to buy this asset when they believe that other currencies are losing their value.

E. DEFLATION PROTECTION

People chose to hoard cash, and the safest place to hold cash is GOLD at any point in time. BLACK fundamentally has the same GOLD characteristics as a store of value, making it a worthy investment to de-risk from the value collapse.

F. EXPONENTIAL REWARDS

With the novel deflationary mechanics, BLACK holders earn passive trustless rewards instantly without staking needs. The benefits are directly proportional to the holding time, very similar to traditional bonds and certificates.

4. TECHNICAL ARCHITECTURE

BLACK PROTOCOL will be implemented as Smart Contracts, including an Open Zeppelin layer to assist with corrections as required. At its core, the contract enforces several rules to maintain price balance when a transaction or sale occurs, as below.

A) If the transaction originates from the Owner's Wallet, the receiver gets 100% of the transaction amount without a fee; else, 10% of the transaction amount will be deducted as tax.

B) The contract will enforce the following when a user initiates a transaction
   i. If the transfer amount is 0.75 % of the sender's balance, the address will be locked for 24 hours.
   ii. If the transfer amount is between 0.5% to 0.75% of the sender's balance, the address will be locked for 12 hours.
   iii. If the transfer amount is below 0.5 % of the sender's balance, No lock occurs
   iv. A sender can make only two transactions within 24 hours

C) The contract also enforces the following when a user receives a transaction.
   i. If a recipient receives above 0.75 % of the wallet balance, the contract will lock the wallet for 24 hours.
   ii. If a recipient receives between 0.5% to 0.75 % of the wallet balance, the contract will lock the wallet for 12 hours.
   iii. If a recipient receives less than 0.5% of the wallet balance, there won't be any lock, and the user can transfer funds instantly.

D) The total supply also plays a critical role during the transaction process
   i) If the total supply is > 500 **MILLION** tokens**,** the protocol removes 4% of the transaction amount forever from the circulation
   ii) If the total supply is between **100 MILLION** and **500 MILLION,** the protocol removes 2% of the transaction amount forever from the circulation

4.1 BLACK PROTOCOL OPERATIONS

PRECHECK

The protocol enforces two levels of precheck restrictions in general. Any point in time,

1) Users can't make more than two transactions from a wallet in 24 hours

2) Users can't transfer more than 1% of the wallet balance in 24 hours

POST CHECK

BLACK achieves consistent purchase power increase after every transaction through the following FIVE novel protocol dynamics

A) Regulates supply after every transaction by burning 4% of the transaction amount forever from circulation.
B) Regulates exponential rewards by distributing 4% of the transaction amount equally among holders.
C) Regulates selling pressure fluctuations by accumulating 1% of the transaction amount in Burn Vault.

D) Regulates exchange liquidity fluctuations by locking 1% of the transaction amount in LP.

4.2 TIME-LOCK DYNAMICS AND ALP DEMONSTRATION

Let's look at how the time-lock mechanics and ALP works by taking the following example.

Sam has 20000 BLACK assets in his wallet

SCENARIO 1:

Sam initiates a transaction to sell 1000 BLACK ASSETS with the exchange

Validation: PRECHECK (FAILED)

The transfer amount is 5% of his wallet balance, exceeding the acceptable 1% daily limit. As a result, the protocol will not allow the transaction

SCENARIO 2:

Sam initiates a transaction to sell 100 BLACK ASSETS with the exchange

Validation: PRECHECK (PASSED)

The transfer amount is 0.5% of the wallet balance and is within the acceptable 1% daily limit. As a result, the platform approves the transaction after performing the following six protocol operations

- Increases the daily transaction counter by 1

- Lock the Wallet for 12 hours since it reaches 50% of the 1% limit

- Accumulates 1% (1 BLACK token) of the transaction amount in Burn Vault.

- Send 1% (1 BLACK token) of the transaction amount to the exchange liquidity pool.

- Burn 4% (4 BLACK tokens) of the transaction amount forever from circulation.

- Distributes 4% (4 BLACK tokens) of the transaction amount as rewards equally among holders.

SCENARIO 3:

Sam initiates a transaction to sell 200 BLACK ASSETS with the exchange

Validation: PRECHECK (PASSED)

The transfer amount is 1% of the wallet balance and is within the acceptable threshold of 1% daily limit. As a result, the platform approves the transaction after performing the following six protocol operations

- Increases the daily transaction counter by 1
- Lock the Wallet for 24 hours since it reaches 100% of the 1% limit
- Accumulates 1% (2 BLACK tokens) of the transaction amount in Burn Vault.
- Send 1% (2 BLACK tokens) of the transaction amount to the exchange liquidity pool.
- Burn 4% (8 BLACK tokens) of the transaction amount forever from circulation.
- Distributes 4% (8 BLACK tokens) of the transaction amount as rewards equally among holders.

SCENARIO 4:

The investor initiates three sell transaction of 50 BLACK ASSETS, each consecutively with the exchange

TRANSACTION 1:

Validation: PRECHECK (PASSED)

- The 1st transfer amount is 0.25% of the wallet balance and is within the acceptable 1% daily threshold. The platform approves the transaction after performing the following five protocol operations

- Increases the daily transaction counter by 1 (Total Transaction in 24 hours = 1)
- Accumulates 1% (0.5 BLACK tokens) of the transaction amount in Burn Vault.
- Send 1% (0.5 BLACK tokens) of the transaction amount to the exchange liquidity pool.
- Burn 4% (2 BLACK tokens) of the transaction amount forever from circulation.
- Distributes 4% (2 BLACK tokens) of the transaction amount as rewards equally among holders.

Note: There won't be any Wallet lock since the % of the transaction amount is less than 0.5%

TRANSACTION 2:

Validation: PRECHECK (PASSED)

- The 2nd transfer amount is 0.25% of the wallet balance and is within the acceptable 1% daily threshold. The platform approves the transaction after performing the following six protocol operations

- Increases the daily transaction counter by 1 (Total Transaction in 24 hours = 2)
- Lock the Wallet for 24 hours because the total transaction reaches the 24-hour threshold
- Accumulates 1% (0.5 BLACK tokens) of the transaction amount in Burn Vault.
- Send 1% (0.5 BLACK tokens) of the transaction amount to the exchange liquidity pool.
- Burn 4% (2 BLACK tokens) of the transaction amount forever from circulation.
- Distributes 4% (2 BLACK tokens) of the transaction amount as rewards equally among holders.

TRANSACTION 3:

Validation: PRECHECK (PASSED)

Though the transfer amount is within the acceptable 1% daily threshold, the wallet is locked after the 2nd transaction; hence the protocol will not approve the 3rd transaction.

4.3: BURN VAULT DEMONSTRATION

The primary objective of the burn vault design is to provide investors a way to cash out their holdings without impacting the market price.

After every transaction, 1% of the transaction amount deducted as tax and accumulated in a separate contract called Burn Vault

The protocol does the following two steps once a transaction amount is deposited

1) Convert the received amount into a stable asset through an exchange
2) Lock the converted stable asset in Burn Vault

With this accumulation process, the assets stored in the Vault always grow and proportional to the daily transaction volume. When a holder decides to cash out and send the token asset to the Burn Vault address, the Burn Vault, in turn, credits his wallet back the proportional amount of stable assets to the Vault price set by the protocol.

VAULT PRICING MODEL:

The Burn Vault Swap price vary every minute and depends on the following two variables

a) Total BLACK Supply (A)
b) Total USD Value locked in Vault (B)

The protocol set a new price dynamically in the Vault using the following formula

Vault Price Per Asset = (Total USD Value locked in Vault)/ (Total BLACK Supply)

For example,

The total circulating supply is 100,000

Total Value Locked is 2000 USD

Then the Vault Price set by the protocol will 0.02 $ (2000 / 100,000)

When a user sends 100 token assets to burn Vault, he will get (100 x 0.02) 2 USD in return

Moreover, the token assets swapped via Burn Vault will be removed permanently from circulation, making the asset more scarce increasing the price further.

BENEFITS AND RESTRICTIONS

1) The protocol also restricts a holder from swapping more than 0.01 % of the total circulating supply using Burn Vault in 30 days.

5. USE CASE – SAFE DEFI HEDGE COLLATERAL

The protocol dynamics are structured in such a way to be a secured store of value in crypto, making it one of the robust candidates for long-term investors. The collateral backing via Burn Vault permanently bonds the value giving a minimum price guarantee to investors, optimizing the risk factor to a bare minimum. The Time-Lock concept further makes it Whale-Proof, avoiding price manipulation and increase the floor price consistently. The deflationary mechanics will drive the price up consistently as the supply becomes scarce and investors earn exponential interest by simply holding the collateral in their wallets. These qualities make black collateral one of the most intelligent investments designed to regulate linear purchase power growth.

6. TOKEN METRICS

Token ticker: BLACK

Total Supply: 1,000,000,000 BLACK

Private round:

25% (250,000,000 BLACK)

[$0.003 per TOKEN]

1-month cliff after TGE, then 25% released every 2 months

Public round:

25% (250,000,000 BLACK)

[$0.0034 per TOKEN] 100% unlocked

Team & Advisors:

20% (200,000,000 BLACK)

3 months cliff after TGE, then 10% released every 2 months

Liquidity:

5% (50,000,000 BLACK)

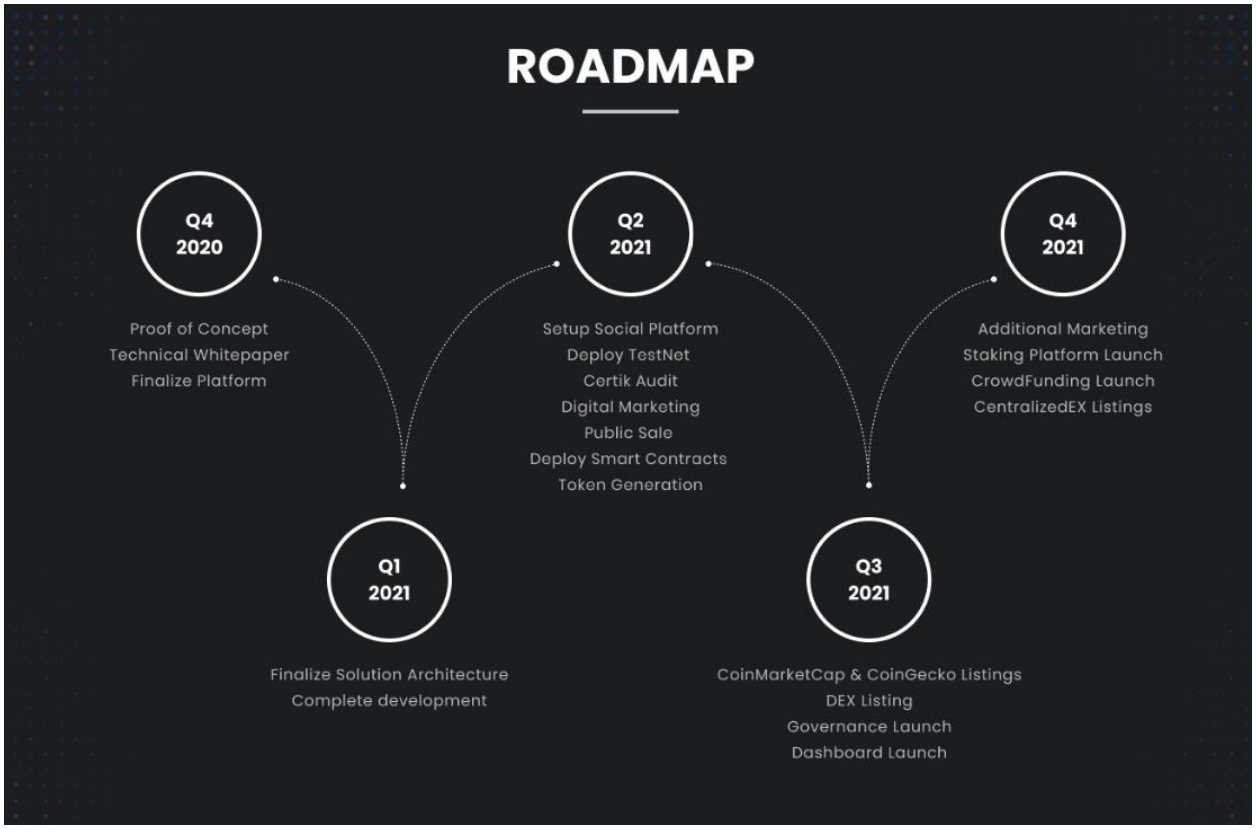
12 months cliff after TGE, then 5% every 3 months

[$0.0035 per TOKEN – Listing price]

Ecosystem Fund:

25% (250,000,000 BLACK)

7. ROADMAP



8. FINAL THOUGHTS

BLACK brings a remarkable degree of innovation in the concept of cryptocurrency mechanics. The protocol fundamentally changes the model from a price-based trading strategy to a time-based strategy that guarantees more reliability and independence from the performance of other cryptocurrencies, including bitcoin. Black's price resistance model opens up many opportunities in DeFi for essential applications that genuinely benefit everyone fairly.

The novel dynamics present an optimistic view for a future with supply-based cryptocurrencies, guaranteeing a more diverse, fair, and independent market. Although the price can be volatile in the short term, its value will increase and stabilizes over the long term. Through the years, similar to GOLD, Black will serve as the best hedge against inflation as an investment well worth considering.

9. REFERENCES

[1] J. Rickards (2015). "The Death of Money" - The Coming Collapse of the International Monetary System

[2] Peter Schiff (2012). "The Real Crash" - America's Coming Bankruptcy.

[3] Niall Ferguson (2013). "The Great Degeneration" - How Institutions and Economics Die.